Problem 1. Unate Covering:

Solve the following covering problem.
NOTE: If Branch and Bound is needed at any point, use the following rules: (i) The next pivoting column should be selected as the one that covers the most rows. (ii) If there are no two independent rows in the cyclic core, the lower bound should be assumed to be equal to 2. **Clearly identify the initial Upper and Lower Bounds and the partial solutions costs and any updates made to the best solution found in your steps.**

	Col1	Col2	Col3	Col4	Col5
Row1	1	1	0	1	0
row2	1	0	1	0	1
row3	1	0	0	1	0
row4	0	1	1	1	0
row5	0	1	0	1	1
row6	1	0	1	1	0

Problem 2:
We can generalize the unate covering problem by associating a **weight** (a nonnegative real number) with each column. The objective of this generalized problem then would be defined as ***minimizing the sum of the weights of the columns selected in the solution***. Below is an example.

	Col1 (weight= 0.05)	Col2 (weight = 110)	Col3 (weight = 7.2)	Col4 (weight= 273.8)
Row1	1		1	1
Row2		1		1
Row3	1		1	1
Row4		1	1	1

How would you define the row dominance rule for this version of the problem?

Problem 3.

Assume gates such as NOT, NAND, NOR, AND, OR have a delay of 1 unit and an XOR or XNOR gate has a delay of 3 units. What is the worst-case delay from any input to any sum output for the 4-bit ripple adder in Figure8-2 ? What is the worst-case delay to the carry output c_4?

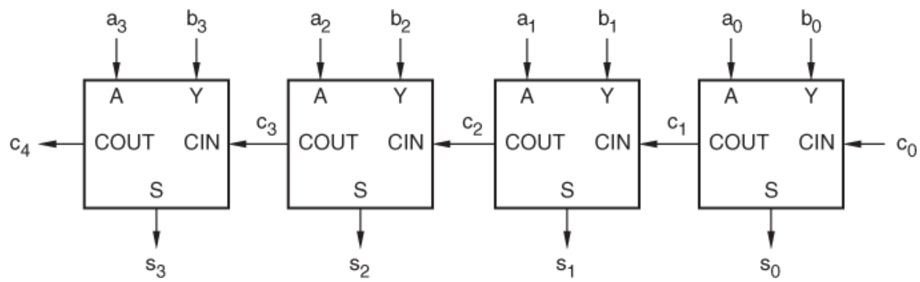

Figure 8-2 A 4-bit ripple adder.